

Blocker Report

Question 1

Objective

To determine whether the current Version 10 rule extraction layer is acceptable in terms of result quality, workflow, runtime, cost, and generalizability.

Attempts

The extraction pipeline was updated to Version 10. Instead of asking an LLM to search the whole bylaw directly, the pipeline first converts the PDF into RAG-like structured blocks, selects rule-relevant blocks using keywords and PDF structure, and then extracts rules only from those selected blocks.

This helps reduce cross-section mistakes compared with the PIBC-style approach. For example, Burnaby R1 does not have a specific tree removal rule, but a broad LLM system can incorrectly apply a P11 tree-removal permission rule to R1. The block-based extraction method reduces this risk by keeping rules tied to their source section and evidence.

Evidence

Final extraction-layer outputs are available for Burnaby and Calgary. These are high-quality extraction outputs, but they are not official verified outputs until they pass the verification layer:

- Burnaby: `code/prototype_pipeline_10/outputs/burnaby/06_rule_extraction_hq/merged_rules_clean.csv`
- Calgary: `code/prototype_pipeline_10/outputs/calgary/06_rule_extraction_hq/merged_rules_clean.csv`

The Calgary bylaw has about 1,035 pages. The current extraction pipeline can extract rules from it in around 3-5 minutes with parallel API calls, with an estimated main extraction cost of about \$0.50.

Diagnosis

The current rule extraction layer appears acceptable for the prototype. Compared with the PIBC-style website approach, the file coverage strategy is similar: bylaw PDFs are downloaded from official government sources, stored locally, and updated when needed. However, the extraction process is more controlled because it narrows the PDF into relevant blocks before rule extraction.

The workflow is designed as generalized PDF processing rather than city-specific hardcoding. Although it has not been tested on every city, it has been exercised on Burnaby, Vancouver, Surrey, and Calgary, and should be adaptable to other municipal bylaw PDFs.

The main remaining limitation is table extraction. Complex tables still require visual language API processing, which is the largest cost component. In difficult cases such as Burnaby tables, one complex table may add about \$0.30-\$0.50 in token cost.

Proposed Next Steps

Complete the latest Vancouver extraction, compare it with Burnaby and Calgary, and continue optimizing complex table extraction.

Decision Ask

Do you agree that the Version 10 rule extraction layer is acceptable for the prototype, with complex table extraction as the main remaining limitation?

Question 2

Objective

To determine whether our verification layer is acceptable as the authority layer for the prototype, meaning extracted rules are not trusted until deterministic verification proves exact source support.

Attempts

Our side of the system treats extraction as candidate generation only. The verification pipeline accepts candidates and evidence units, checks each rule against source-backed evidence, and then separates outputs into verified, review-needed, rejected, or not-used buckets.

The current product path is native M4. It uses full-bylaw discovery, RAG/evidence packs, native extraction, source repair, deterministic verification, and a verified-only GIS contract. The trust boundary is:

```
Extraction proposes.  
Source repair restores evidence context.  
Deterministic verification decides.  
GIS consumes only verified rules.
```

Uncertain rules are not promoted. They stay in `review_needed.json`, and advisory review tools such as semantic review, evidence repair, and review routing do not become authority.

Evidence

Current MVP report:

```
outputs/mvp_verification/mvp_report.md
overall_status = mvp_safety_ready
current_path_version = native_m4
current_false_verified_total = 0
```

Current native M4 benchmark rows:

```
Burnaby R1: 101 candidates, 84 verified, 17 review, 0 rejected, 0 false verified
Vancouver RS: 32 candidates, 12 verified, 11 review, 9 rejected, 0 false verified
Calgary R-CG: 306 candidates, 11 verified, 43 review, 35 rejected, 217 not used, 0 false verified
```

Burnaby native M4 benchmark:

```
Gold rules: 40
Candidate rules: 101
Verified rules: 84
Review rules: 17
Rejected rules: 0
Verified precision: 1.00
Verified-or-review recall: 1.00
False verified rules: 0
Source support failures: 0
```

Source-PDF audit:

```
Burnaby R1: 84 verified rules checked, 0 failures
Vancouver RS: 12 verified rules checked, 0 failures
Calgary R-CG: 11 verified rules checked against the full 1,053-page bylaw, 0 failures
```

Diagnosis

The verification layer is acceptable as the safety authority for the prototype. It satisfies the most important safety condition: no unsupported rule is promoted as verified in the benchmarked product path.

The main remaining limitation is not unsafe verification. The limitation is review efficiency and evidence quality. Burnaby native M4 still has 17 rules in review, mostly because text candidates need consensus, applies-to scope is missing, conditions are incomplete, or exception language requires manual legal review. Evidence quality reports also show missing bounding boxes and source context issues for table-derived evidence, which makes review harder even when the verifier remains safe.

Proposed Next Steps

Keep the deterministic verifier as the authority layer. Do not relax verification to increase the verified count.

Next, improve evidence metadata and review efficiency:

- add better row, column, heading, page, and bounding-box context for table evidence;
- reduce review cases only when benchmark scores prove the change is safer or equally safe;

- keep exception, unsupported scope, and condition gaps in review until exact source support exists;
- continue validating changes against Burnaby plus holdout cities.

Decision Ask

Do you agree that the native M4 verification layer is acceptable as the prototype authority layer, with review efficiency and table-evidence metadata as the main remaining limitations?

Question 3

Objective

To determine whether our downstream GIS and proposal-compliance layer is acceptable for the prototype, as long as it is presented as a verified-only tri-state checker rather than a final legal permitting engine.

Attempts

Our downstream layer consumes only verified rules. The GIS handoff is `gis_rule_contract.json`, which excludes review-needed and rejected candidates. The proposal checker in `compliance.py` evaluates proposal fields against verified rules and returns one of three decisions:

```
approved
rejected
needs_review
```

The safety rule is conservative: if a proposal field is not covered by a verified rule, the checker returns `needs_review` instead of approving it.

Evidence

Burnaby native M4 GIS contract:

```
outputs/m4_runs/burnaby_r1/google_gemini_2_5_flash_lite/gis_rule_contract.json
schema_version = 1.1
rules exported = 84
source = verified rules only
```

Burnaby native M4 proposal benchmark:

```
Proposal cases: 10
Decision accuracy: 1.00
Case accuracy including expected fields: 1.00
False approvals: 0
False rejections: 0
Field expectation mismatches: 0
Needs-review decisions: 2
```

The proposal benchmark includes approval, rejection, and review cases. The current M4 run passes all cases, including:

```
approved_verified_core_flat -> approved
approved_verified_core_sloping_exact -> approved
reject_flat_roof_height -> rejected
reject_front_rear_separation -> rejected
reject_accessory_rear_yard -> rejected
needs_review_rear_principal_rear_yard -> needs_review
needs_review_heritage_coverage -> needs_review
approve_all_verified_setbacks_and_separations -> approved
```

Diagnosis

The downstream layer is acceptable for the prototype if its scope is stated carefully. It can demonstrate safe GIS and proposal decisions for fields that are covered by verified rules. It should not be described as a complete permitting system because the benchmark only covers a bounded set of proposal fields and rule families.

The main remaining limitation is coverage of real-world proposal inputs. Additional geometry, parcel context, lot attributes, zoning overlays, and exception handling would be needed before this could become a broader municipal approval tool. For the capstone prototype, the important result is that the checker can approve or reject when the verified-only contract is sufficient, and can return `needs_review` when it is not.

Proposed Next Steps

Keep GIS and proposal decisions downstream of verified rules only.

Next, expand the benchmark and contract gradually:

- add more proposal cases for edge values, missing fields, heritage conditions, fire access, street frontage, and table-derived scope;
- add adversarial proposal cases that should never be approved;
- keep field-level expectation checks, not only final decision checks;
- expose review-needed proposal fields clearly in the GIS or dashboard handoff.

Decision Ask

Do you agree that the GIS/proposal-compliance layer is acceptable for the prototype as a verified-only tri-state checker, with broader real-world proposal coverage as the main remaining limitation?